Algebraic Fractions - Mark scheme

Algebra · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

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Question 1 *Practice question written for this worksheet (not from a past paper)*

Simplify $\frac{12x^2}{4x}$

Divide 12 by 4; $x^2 \div x = x$

Answer: 3x

Question 2 *Practice question written for this worksheet (not from a past paper)*

Simplify $\frac{x^2 + 3x}{x}$

Factorise the top: $\frac{x(x+3)}{x}$

Answer: x + 3

Question 3 *Practice question written for this worksheet (not from a past paper)*

Simplify $\frac{5x + 10}{x + 2}$

$\frac{5(x+2)}{x+2}$

Answer: 5

Question 4 *Practice question written for this worksheet (not from a past paper)*

Write $\frac{x+1}{2} + \frac{x+2}{3}$ as a single fraction.

Common denominator 6: $\frac{3(x+1)+2(x+2)}{6}$

$\frac{3x+3+2x+4}{6}$

Answer: (5x + 7)/6

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Question 5 *Practice question written for this worksheet (not from a past paper)*

Write $\frac{2x-1}{3} - \frac{x+4}{6}$ as a single fraction in its simplest form.

$\frac{2(2x-1)-(x+4)}{6} = \frac{4x-2-x-4}{6} = \frac{3x-6}{6}$

Divide top and bottom by 3

Answer: (x - 2)/2

Question 6 *Practice question written for this worksheet (not from a past paper)*

Write $\frac{1}{x} + \frac{1}{y}$ as a single fraction.

Common denominator xy: $\frac{y}{xy} + \frac{x}{xy}$

Answer: (x + y)/(xy)

Question 7 *Practice question written for this worksheet (not from a past paper)*

Simplify $\frac{6x^2y}{9xy^2}$

Divide by 3, then cancel x and y

Answer: 2x/(3y)

Question 8 *Practice question written for this worksheet (not from a past paper)*

Simplify $\frac{x^2-y^2}{3x+3y}$

Top: $(x-y)(x+y)$; bottom: $3(x+y)$

Cancel $(x+y)$

Answer: (x - y)/3

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Question 9

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2023 · Question 11

Q	Working	Answer	Mark	Notes
	$\frac{9x}{12x}+\frac{2(5-x)}{12x}$ oe or or $\frac{3(6x)}{24x}+\frac{4(5-x)}{24x}$ oe or $\frac{3(6x)}{4(6x)}+\frac{4(5-x)}{4(6x)}$ oe or or $\frac{18x}{24x}+\frac{20-4x}{24x}$ oe or $\frac{3\times3x+2(5-x)}{12x}$ oe		3	M1 for two correct fractions with common denominator with the intention to add **or** a single correct fraction
	$\frac{9x+10-2x}{12x}$ oe or $\frac{18x+20-4x}{24x}$ oe or $\frac{14x+20}{24x}$ oe or			M1 for a correct numerator over a single denominator with brackets expanded and correct signs Allow $\frac{7x}{12x}+\frac{10}{12x}$
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	$\frac{7x+10}{12x}$		A1 oe $\frac{10+7x}{12x}$
				Total 3 marks

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Question 10

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2025 · Question 12

Q	Working	Answer	Mark	Notes
	eg $\frac{5(6x)}{24x}+\frac{4(x-3)}{24x}$ oe **or** $\frac{5(6x)}{4(6x)}+\frac{4(x-3)}{4(6x)}$ oe **or** $\frac{30x}{24x}+\frac{4(x-3)}{24x}$ oe **or** $\frac{30x+4(x-3)}{24x}$ oe **or** $\frac{15x}{12x}+\frac{2(x-3)}{12x}$ oe **or** $\frac{15x+2(x-3)}{12x}$ oe		3	M1 for two correct fractions with common denominator **or** a single correct fraction
	eg $\frac{30x+4x-12}{24x}$ oe **or** $\frac{30x}{24x}+\frac{4x-12}{24x}$ oe **or** $\frac{30x}{24x}+\frac{4x}{24x}-\frac{12}{24x}$ oe **or** $\frac{34x}{24x}-\frac{12}{24x}$ oe **or** $\frac{34x-12}{24x}$ oe **or** $\frac{15x+2x-6}{12x}$ oe			M1 for correct fraction(s) with bracket(s) expanded correctly
	Correct answer scores full marks (unless from obvious incorrect working)	$\frac{17x-6}{12x}$		A1 oe but must be simplified eg $\frac{-6+17x}{12x}$ do not ISW incorrect simplification eg $\frac{17x-6}{12x}=\frac{11}{12}$ is M2A0
				Total 3 marks

Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2016 · Question 14

14	$\frac{(x-5)(x+5)}{(2x+1)(x-5)}$	$\frac{x+5}{2x+1}$	3	M1 For $(x+5)(x-5)$ M1(indep) For $(2x+1)(x-5)$ or $2(x+0.5)(x-5)$ or $2(2x+1)(0.5x-2.5)$ A1 cao No ISW
				Total 3 marks

Question 12

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2013 · Question 17

17.	$\frac{(2x-5)(2x+5)}{(2x+5)(3x-1)}$	$\frac{(2x-5)}{(3x-1)}$	3	M2 If not M2 then M1 for numerator or denominator correct A1
				Total 3 marks

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Question 13

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2026 · Question 18

Question	Working	Answer	Mark	Notes
	$(2x-3)(2x+3)$ **or** $(2x+3)(x-4)$		3	M1 for correct factorisation of either the numerator **or** the denominator
	$(2x-3)(2x+3)$ **and** $(2x+3)(x-4)$			M1 for correct factorisation of the numerator **and** denominator
	Working required	$\frac{2x-3}{x-4}$		A1 (dep on M2) fully correct factorising and cancelling No ISW
				Total 3 marks

Question 14

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2012 · Question 20

20.	$\frac{4(2-x)+3x}{x(2-x)}$ oe $\frac{8-4x+3x}{x(2-x)}$	$\frac{8-x}{x(2-x)}$	3	M1 M1 A1 Accept $\frac{8-x}{2x-x^2}$ Single fraction needed as final answer.
				Total 3 marks

Question 15

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2019 · Question 22

22		$x(2x + 5)(3x - 1)$ or $(2x + 5)(3x^2 - x)$ or $(2x + 5)(2x - 5)$ oe			M1 for a correct factorisation of the numerator into 2 or 3 factors where one of the factors must be $(2x + 5)$ **or** denominator into 2 brackets where one of the factors must be $(2x + 5)$
		$x(2x + 5)(3x - 1)$ or $(2x + 5)(3x^2 - x)$ **and** $(2x + 5)(2x - 5)$ oe			M1 for a correct factorisation of the numerator into 2 or 3 factors where one of the factors must be $(2x + 5)$ **and** denominator into 2 brackets where one of the factors must be $(2x + 5)$
			$\frac{x(3x-1)}{2x-5}$	3	A1 accept $\frac{3x^2-x}{2x-5}$ oe Do not ISW
					Total 3 marks

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Question 16

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2022 · Question 23

23	eg $\frac{20}{x^2-36}-\frac{2(x+6)}{x^2-36}$ oe **or** $\frac{20}{(x-6)(x+6)}-\frac{2(x+6)}{(x-6)(x+6)}$ oe **or** $\frac{20(x-6)}{(x^2-36)(x-6)}-\frac{2(x+6)(x-6)}{(x^2-36)(x-6)}$ or $\frac{20-2(x+6)}{(x^2-36)(4-x)}$ oe		3	M1	for writing the first two fractions with a common denominator (may be a single denominator) or multiplying both fractions by $\frac{1}{4-x}$ and writing over a common denominator
	eg $\frac{8-2x}{x^2-36}\times\frac{1}{4-x}$ **or** $\frac{8-2x}{(x-6)(x+6)}\times\frac{1}{4-x}$ **or** $\frac{20x-2x^2-48}{(x^2-36)(x-6)}\times\frac{1}{4-x}$ oe $\frac{8-2x}{(x^2-36)(4-x)}$ oe			M1	for simplifying first 2 fractions to a single fraction and expanding and simplifying numerator – must be correct, and showing intention to multiply by $\frac{1}{4-x}$ or expanding the numerator of the full solution and writing as a single fraction
		$\frac{2}{x^2-36}$		A1	oe eg $\frac{2}{(x-6)(x+6)}$
					Total 3 marks

Question 17

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2021 · Question 21

21		$\frac{(5x-8)(5x+8)}{(5x+2)(x-3)}\times\frac{(x-5)(x-3)}{5x+8}$ or eg $\frac{(5x-8)(x-5)}{(5x+2)}(-(x-7))$		4	M2 For factorising at least 2 of the quadratics correctly – could be implied by 2 factors cancelled correctly (M1 For factorising at least 1 of the 3 quadratics correctly)
		$\frac{(5x-8)(x-5)-(x-7)(5x+2)}{5x+2}$ oe or $\frac{5x^2-25x-8x+40-(5x^2-35x+2x-14)}{5x+2}$ oe or $\frac{(25x^2-64)(x^2-8x+15)-(x-7)(5x^2-13x-6)(5x+8)}{(5x^2-13x-6)(5x+8)}$ oe or $\frac{(5x-8)(x^2-8x+15)-(x-7)(5x+2)(x-3)}{(5x+2)(x-3)}$ oe or $\frac{(25x^2-64)(x-5)-(x-7)(5x+2)(5x+8)}{(5x+2)(5x+8)}$ oe			M1 (indep (ft if M2 awarded)) For writing the fractions over a common denominator with or without brackets removed – need not be in simplest form Could be written as 2 separate fractions over a common denominator
			$\frac{54}{5x+2}$		A1 dep on M3
					Total 4 marks

Question 18

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2018 · Question 22

22	$\frac{(2x+5)(2x-5)}{(5x+7)(x-1)}\times\frac{2(2x-5)-3(x-3)}{(x-3)(2x-5)}$ $\frac{(2x+5)(2x-5)}{(5x+7)(x-1)}\times\frac{x-1}{(x-3)(2x-5)}$ $\frac{2x+5}{(5x+7)(x-3)}$	$\frac{2x+5}{(5x+7)(x-3)}$	4	M1	For $4x^2 - 25 = (2x + 5)(2x - 5)$ or $5x^2 + 2x - 7 = (5x + 7)(x - 1)$
				M1	$\frac{2}{x-3}-\frac{3}{2x-5}=\frac{2(2x-5)-3(x-3)}{(x-3)(2x-5)}$ oe
				M1	$\frac{(2x+5)(2x-5)}{(5x+7)(x-1)}\times\frac{x-1}{(x-3)(2x-5)}$ oe may be partially simplified
				A1	Denominator may be expanded eg $\frac{2x+5}{5x^2-8x-21}$ isw for incorrect denominator expansion
					Total 4 marks

Question 19

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2025 · Question 23

23	$\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \div \frac{x(x+5)}{10x(x+6)} \ (= p)$ or $\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \times \frac{10x(x+6)}{x(x+5)} \ (= p)$ or 2 from $4(x-6)(x+5)$ or $5(x-6)(x+6)$ or $x(x+5)$ or $10x(x+6)$		4	M1	for factorising 2 or 3 of the quadratics fully – could be implied by 2 factors cancelled correctly NB factors must be in the form $(ax + b)$ **NB Substitution of values of x into the given equation is not an acceptable algebraic method**
	$\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \div \frac{x(x+5)}{10x(x+6)} \ (= p)$ or $\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \times \frac{10x(x+6)}{x(x+5)} \ (= p)$ or $4(x-6)(x+5)$ and $5(x-6)(x+6)$ and $x(x+5)$ and $10x(x+6)$			M1	for factorising **all** of the quadratics fully – could be implied by 2 factors cancelled correctly NB factors must be in the form $(ax + b)$
	$\frac{4x^2-4x-120}{5x^2-180} \times \frac{10x^2+60x}{x^2+5x} \ (= p)$			M1	for inverting the 2nd fraction (this mark can be awarded at any time and may be awarded with incorrect factorisation if meaning is clear)
	Working required	8		A1	oe dep on M3
					Total 4 marks
ALT	$\frac{40x^4+200x^3-1440x^2-7200x}{5x^4+25x^3-180x^2-900x}$			M3	for a correct expression (no errors)
		8		A1	oe dep on M3

Question 20

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2024 · Question 23

23	$16x^2-36=(4x-6)(4x+6)$ $[=4(2x+3)(2x-3)]$ oe or $16x^2-36=(8x-12)(2x+3)$ $[=4(2x+3)(2x-3)]$ oe		4	M1indep	
	$2x^2+7x+6=(2x+3)(x+2)$ **and** $x^2-5x-14=(x-7)(x+2)$ [We will make an exception of $2x^2+7x+6=(4x+6)(0.5x+1)$ as this then cancels with $(4x + 6)$]			M1indep	[NB: the two fractions when divided and cancelled give an answer of $4(2x – 3)$ or $2(4x – 6)$ or $8x – 12$ (any one of these gain 2 marks)] M2 for any fraction with completely simplified non-linear numerator and non-linear denominator that will cancel to $–19$
	$4(2x-3)-(7+8x)(=n)$ [allow invisible brackets ie $8x – 12 – 7 + 8x$] or an expression that clearly shows the numerator is – 19 times the denominator eg $\frac{-38x-57}{2x+3}(=n)$			M1	a linear expression that should give the correct value for n (this mark implies previous M marks as not all factorising is necessary)
	Working required	$–19$		A1	dep on M2
					Total 4 marks

Question 21

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2015 · Question 23

23	$x^2-4=(x+2)(x-2)$		4	B1	independent
	$[5-](x+2)\times\frac{(x-3)}{(x^2-4)}$ or $[5-](x+2)\times\frac{(x-3)}{(x-2)(x+2)}$			M1	for dealing with division of $(x+2)$ by $\frac{x^2-4}{x-3}$
	$\frac{5(x-2)}{(x-2)}-\frac{(x-3)}{(x-2)}$ or $\frac{5(x-2)(x+2)}{(x-2)(x+2)}-(x+2)\times\frac{(x-3)}{(x-2)(x+2)}$ or $\frac{5(x^2-4)}{x^2-4}-(x+2)\times\frac{(x-3)}{x^2-4}$			M1	For two correct fractions with a common denominator or a correct single fraction
		$\frac{4x-7}{x-2}$		A1	from fully correct algebra
					Total 4 marks

Question 22

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2020 · Question 24

24	$\frac{4(2x-3)-3(2x-5)}{(2x-5)(2x-3)}$ or $\frac{8x-12-6x+15}{(2x-5)(2x-3)}$ oe		4	M1	Writing 1st fraction as a fraction over a common denominator (can be 2 separate fractions)
	$x(3-2x)(3+2x)$ or $(3x-1)(2x-5)$			M1	Complete factorisation of numerator or denominator of 2nd fraction
	$\frac{2x+3}{(2x-5)(2x-3)}\times\frac{(3x-1)(2x-5)}{x(3-2x)(3+2x)}$			M1	may be partially simplified
		$\frac{3x-1}{x(2x-3)(3-2x)}$		A1	e.g. $\frac{3x-1}{x(2x-3)(3-2x)}$ or $\frac{1-3x}{x(2x-3)^2}$ or $\frac{3x-1}{x(12x-9-4x^2)}$ or $\frac{3x-1}{(12x^2-9x-4x^3)}$ oe isw for incorrect denominator expansion
					Total 4 marks

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Question 23

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2024 · Question 26

26	$(3x-5)(x+2)$	$\frac{4(3x^2+x-10)-(3x-5)(4x-1)}{3x^2+x-10}$ oe or $\frac{27x-45}{3x^2+x-10}$		4	M1	for correctly factorising $3x^2+x-10$ to give $(3x-5)(x+2)$ (may be seen later on in working) **OR** combining 2 fractions into a correct single fraction
	$\frac{4x-1}{x+2}$ implies first M1	$\frac{4(3x^2+x-10)-(3x-5)(4x-1)}{(3x-5)(x+2)}$ or $\frac{27x-45}{(3x-5)(x+2)}$			M1	for inverting and cancelling giving a correct fraction **OR** for a correct single fraction where the denominator is factorised
	$\frac{4(x+2)-(4x-1)}{x+2}$ or $\frac{4x+8-4x+1}{x+2}$ or $\frac{4(x+2)}{x+2}-\frac{4x-1}{x+2}$ or $\frac{4x+8}{x+2}-\frac{4x-1}{x+2}$	$\frac{9(3x-5)}{(3x-5)(x+2)}$			M1	for a correct single fraction or two correct fractions with a common denominator **OR** for a correct fully factorised single fraction
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)		$\frac{9}{x+2}$		A1	
						Total 4 marks

Question 24

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2023 · Question 26

26	eg $\frac{(4x+3)(x-5)}{2x-1}\times\frac{(2x-1)(x-3)}{(x+5)(x-5)}$ **or** eg $\frac{(4x+3)(x-3)}{x+5}(+(29-4x))$		4	M2	for factorising at least 2 of the quadratics correctly – could be implied by 2 factors cancelled correctly (M1 for factorising at least 1 of the 3 quadratics correctly)
	eg $\frac{(4x+3)(x-3)+(29-4x)(x+5)}{x+5}$ oe **or** eg $\frac{4x^2-9x-9+145+9x-4x^2}{x+5}$ oe			M1	for writing the correct fractions over a common denominator of $(x+5)$ with or without brackets removed – need not be in simplest form. Could be written as 2 separate fractions.
	Correct answer scores full marks (unless from obvious incorrect working)	$\frac{136}{x+5}$		A1	
					Total 4 marks

Question 25

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2018 · Question 15

15	a	E.g. $\left(\frac{y^8}{256x^{20}}\right)^{\frac{1}{4}}$ **or** $\left(\frac{4x^5}{y^2}\right)^{-1}$ **or** $\frac{x^{-5}}{4y^{-2}}$ **or** $\frac{\frac{1}{4}x^{-5}}{y^{-2}}$ **or** $k\frac{y^a}{x^b}$ **or** $\frac{ky^a}{x^b}$ with 2 of $k=\frac{1}{4}$ oe , $a=2$, $b=5$ **or** $\frac{y^a}{mx^b}$ with 2 of $m=4$, $a=2$, $b=5$	$\frac{y^2}{4x^5}$	2	M1	for a correct first step leading to a correct partially simplified expression
					A1	for $\frac{y^2}{4x^5}$ **or** $\frac{\frac{1}{4}y^2}{x^5}$ **or** $0.25\frac{y^2}{x^5}$ **or** $0.25y^2x^{-5}$
	b	$\frac{1}{(3x-5)(3x+5)}-\frac{1}{2(3x+5)}$ E.g. $\frac{2}{2(3x-5)(3x+5)}-\frac{1(3x-5)}{2(3x-5)(3x+5)}$ **or** $\frac{6x+10}{(9x^2-25)(6x+10)}-\frac{9x^2-25}{(9x^2-25)(6x+10)}$	$\frac{7-3x}{2(3x-5)(3x+5)}$	3	M1	indep for $(3x+5)(3x-5)$
					M1	for two correct fractions with a common denominator if there is any expansion at this stage then it must be correct
					A1	accept equivalents eg. $\frac{7-3x}{18x^2-50}$
		Alternative scheme $\frac{6x+10}{(9x^2-25)(6x+10)}-\frac{9x^2-25}{(9x^2-25)(6x+10)}$	$\frac{7-3x}{2(3x-5)(3x+5)}$	3	M1	for two correct fractions with a common denominator
		$\frac{(7-3x)(3x+5)}{(9x^2-25)(6x+10)}$			M1	Numerator expanded and then factorised correctly
					A1	accept equivalents

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Question 26

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2020 · Question 21

Q	Working	Answer	Mark	Notes
	$\left(\frac{9x^2-4}{3x^2-13x-10}=\right)\ \frac{(3x+2)(3x-2)}{(3x+2)(x-5)}$			M1 for either $(3x+2)(3x-2)$ **or** $(3x+2)(x-5)$ M2 for $\frac{9x^2-4}{(9x^2-4)(x-5)}=\frac{1}{(x-5)}$
	$\left(\frac{9x^2-4}{3x^2-13x-10}=\right)\ \frac{(3x+2)(3x-2)}{(3x+2)(x-5)}$			M1 for $(3x+2)(3x-2)$ **and** $(3x+2)(x-5)$
	E.g. of denominators $(3x-2)(3x^2-13x-10)(x-1)$ **or** $(3x-2)(3x+2)(x-5)(x-1)$ **or** $9x^4-54x^3+41x^2+24x-20$ **or** $(3x+2)(x-5)(x-1)$ **or** $3x^3-16x^2+3x+10$ **or** $(3x-2)(x-5)(x-1)$ **or** $3x^3-20x^2+27x-10$ **or** $(x-5)(x-1)$ **or** x^2-6x+5			M1 (indep) ft their fractions for use of a correct common denominator for 2 fractions with **algebraic** denominators NB: fractions need not be simplified
	$\frac{x-1-7(x-5)}{(x-5)(x-1)}$ **or** $\frac{x-1-7x+35}{(x-5)(x-1)}$ **or** $\frac{x-1-7(x-5)}{x^2-6x+5}$ **or** $\frac{x-1-7x+35}{x^2-6x+5}$ oe			M1 for a **correct** fraction with a **correct quadratic** denominator – may or may not be expanded which leads to a correct answer
		$\frac{2(17-3x)}{(x-5)(x-1)}$	5	A1 accept $\frac{34-6x}{(x-5)(x-1)}$ oe; if denominator is expanded then it must be correct
				Total 5 marks

CHALLENGE QUESTIONS

Question 27

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2017 · Question 22

22	E.g. $\frac{3}{2(x+6)}-\frac{x-15}{(x-8)(x+6)}$ or $\frac{3}{2x+12}-\frac{x-15}{(x-8)(x+6)}$			M1 $x^2-2x-48$ correctly factorised NB : May be seen at a later stage
	E.g. $\frac{3(x-8)-2(x-15)}{2(x-8)(x+6)}$ **or** $\frac{3(x-8)}{2(x-8)(x+6)}-\frac{2(x-15)}{2(x-8)(x+6)}$			M1 for a correct common denominator with numerators correct This may be a single fraction or two fractions; denominators may be expanded – if so, must be correct.
	$\frac{3x-24-2x+30}{2(x-8)(x+6)}$			M1 for a correct single fraction with brackets in numerator removed correctly; denominators may be expanded – if so, must be correct.
	$\frac{x+6}{2(x-8)(x+6)}$			M1 for a correct single fraction with the numerator simplified; denominators may expanded – if so, must be correct.
		$\frac{1}{2(x-8)}$	5	A1 dep on at least M2 for $\frac{1}{2(x-8)}$ or $\frac{1}{2x-16}$ or $\frac{-1}{16-2x}$ or $\frac{-1}{2(8-x)}$
				Total 5 marks

Question 28

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2023 · Question 23

	$(x+2)(x-2)$ oe or $(4x+1)(x-2)$ oe		4	M1	for complete factorisation of x^2-4 **or** $4x^2-7x-2$ Each factor must be in the form $(ax \pm b)$ where a and b are integers
	$(x+2)(x-2)\times\frac{x}{(4x+1)(x-2)}$ or $\frac{x(x+2)(x-2)}{(4x+1)(x-2)}$ or $\frac{x(x+2)}{(4x+1)}$			M1	for complete factorisation of $4x^2-7x-2$ **and** x^2-4 and inverting and intention to multiply
	$\frac{x(x+2)-2x(4x+1)}{(4x+1)}$ or $\frac{x^2+2x-8x^2-2x}{(4x+1)}$ or $\frac{x(x+2)}{(4x+1)}-\frac{2x(4x+1)}{(4x+1)}$ or $\frac{x^2+2x}{(4x+1)}-\frac{8x^2+2x}{(4x+1)}$			M1	for a correct single fraction following correct cancellation or for two correct fractions with common denominator following correct cancellation
	Correct answer scores full marks (unless from obvious incorrect working)	$\frac{-7x^2}{4x+1}$		A1	oe but must be in form $\frac{ax^2}{bx+c}$ where a, b and c are integers.
					Total 4 marks

ALT	$\frac{-7x^3+14x^2}{4x^2-7x-2}$ oe		4	M1	for a correct single fraction
	$\frac{-7x^2(x-2)}{(4x+1)(x-2)}$ oe			M1	for complete factorisation of $-7x^3+14x^2$ **or** $4x^2-7x-2$ Each factor must be in the form $(ax \pm b)$
	$\frac{-7x^2(x-2)}{(4x+1)(x-2)}$ oe			M1	for complete factorisation of $-7x^3+14x^2$ **and** $4x^2-7x-2$ Each factor must be in the form $(ax \pm b)$
	Correct answer scores full marks (unless from obvious incorrect working)	$\frac{-7x^2}{4x+1}$		A1	oe but must be in form $\frac{ax^2}{bx+c}$ where a, b and c are integers.
					Total 4 marks

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Question 29

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2022 · Question 11

Q	Working	Answer	Mark	Notes
(a)	$(3x\pm 2y)(3x\pm 2y)$ or $(3x)^2-(2y)^2$		2	M1
		$(3x+2y)(3x-2y)$		A1
(b)	$\frac{7(4x)}{32x}-\frac{8(x+3)}{32x}$ oe or $\frac{7(4x)}{8(4x)}-\frac{8(x+3)}{8(4x)}$ oe or $\frac{28x}{32x}-\frac{8(x+3)}{32x}$ oe or $\frac{28x}{32x}-\frac{8x+24}{32x}$ oe or $\frac{28x-8(x+3)}{32x}$ oe or $\frac{7x}{8x}-\frac{2(x+3)}{8x}$ oe or $\frac{7x-2(x+3)}{8x}$ oe		3	M1 for two correct fractions with common denominator **or** a single correct fraction
	$\frac{28x-8x-24}{32x}$ oe or $\frac{20x-24}{32x}$ oe or $\frac{7x-2x-6}{8x}$ oe or $\frac{20x}{32x}-\frac{24}{32x}$ oe or $\frac{28x}{32x}-\frac{8x}{32x}-\frac{24}{32x}$ oe			M1 for correct fraction(s) with bracket(s) expanded **and** dealing with the negative signs
		$\frac{5x-6}{8x}$		A1 or $\frac{-6+5x}{8x}$
				Total 5 marks

Question 30

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2026 · Question 16

Q	Working	Answer	Mark	Notes
(a)	eg $\frac{5\times x}{8\times x}-\frac{2(x+2)}{2\times 4x}$ **or** **or** $\frac{5x}{8x}-\frac{2x+4}{8x}$ **or** $\frac{5x-2(x+2)}{8x}$ **or** $\frac{4\times 5x}{4\times 8x}-\frac{8(x+2)}{8\times 4x}$ **or** $\frac{20x}{32x}-\frac{8x+16}{32x}$ **or** $\frac{20x-8(x+2)}{32x}$ oe		3	M1 for two correct fractions with a common denominator **or** a single correct fraction condone M1 for $\frac{5x-2x+4}{8x}$ or $\frac{3x+4}{8x}$ or $\frac{20x-8x+16}{32x}$ or $\frac{12x+16}{32x}$
	eg $\frac{5x-2x-4}{8x}$ **or**			M1 for a correct single fraction with bracket expanded

	$\frac{20x-8x-16}{32x}$ **or** $\frac{12x-16}{32x}$ oe				with bracket expanded
	Correct answer scores full marks (unless from obvious incorrect working)	$\frac{3x-4}{8x}$		A1	oe but must be in simplest form eg $\frac{-4+3x}{8x}$ ISW correct separating into 2 fractions eg $\frac{3x-4}{8x}=\frac{3}{8}-\frac{1}{2x}$ Do not ISW incorrect simplification eg $\frac{3x-4}{8x}=-\frac{1}{8}$ is M2A0

(b)	eg $3x(2x-1)=6x^2-3x$ **or** $3x(x+4)=3x^2+12x$ **or** $(2x-1)(x+4)=2x^2+8x-x-4$ $(=2x^2+7x-4)$		3	M1 for sight of $6x^2-3x$ or $3x^2+12x$ or $2x^2+8x-x-4$ or $2x^2+7x-4$ or for an expansion with only one error	M2 for 3 (out of a maximum of 4) of $6x^3+24x^2-3x^2-12x$ this can be implied by $6x^3+21x^2...$ or $...+21x^2-12x$ (M1 for 2 correct out of a maximum of 4)
	eg $[(6x^2-3x)(x+4)=]\ 6x^3+24x^2-3x^2-12x$ **or** $[(3x^2+12x)(2x-1)=]\ 6x^3-3x^2+24x^2-12x$ **or** $[3x(2x^2+8x-x-4)=]\ 6x^3+24x^2-3x^2-12x$ **or** $[3x(2x^2+7x-4)=]\ 6x^3+21x^2-12x$			M1 ft dep on M1 and a quadratic expression Allow one further error	
	Correct answer scores full marks (unless from obvious incorrect working)	$6x^3+21x^2-12x$		A1 cao (terms may be in any order but must be simplified) ISW correct factorisation eg $3(2x^3+7x^2-4x)$ Do not ISW incorrect simplification eg $6x^3+21x^2-12x=2x^3+7x^2-4x$	
					Total 6 marks